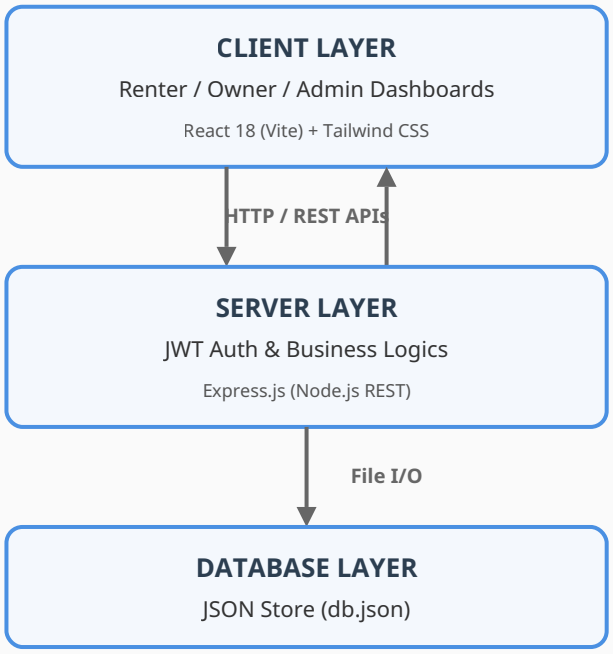
Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	July 2026
Project Name	RentEase Platform
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per table 1 & table 2.



Guidelines Met:

- **Application Logic:** Express.js routing and React SPA logic defined.
- **Infrastructure Demarcation:** Client-Server unified port mapping (Local/Cloud compatible).
- **External Interfaces:** REST API endpoints serving client interface.
- **Data Storage:** Lightweight JSON Store (db.json) for rapid transaction handling.

Table-1 : Components & technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application (Renter Dashboard, Owner Dashboard, Admin Panel)	React 18 (Vite), Tailwind CSS
2.	Application Logic-1	Logic for processing application routing and REST API endpoints	Express.js, Node.js
3.	Application Logic-2	Logic for core business workflows (Lease Scheduler, Listing Composer)	JavaScript (Node.js)
4.	Database	Lightweight transactional data type modeling Users, Properties, and Bookings	JSON Store (db.json)
5.	Authentication / Security	Cryptographically signed payload for stateful session security	JSON Web Tokens (JWT)

S.No	Component	Description	Technology
6.	Infrastructure Pipeline	Application Deployment / Bundling Server assets	esbuild (Unified Server-Asset Pipeline)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used for client and server architecture	React, Tailwind CSS, Express.js, Vite
2.	Security Implementations	List all the security / access controls implemented (RBAC, Session state)	JWT Authentication, Express Authorization Middleware
3.	Scalable Architecture	Justify the scalability of architecture (Client-Server SPA logic)	Single Page Application (SPA), RESTful APIs
4.	Availability & Performance	Design consideration for performance (Unified Port Mapping, Hot Module Replacement)	Vite Pre-building, Express Static Serving (/dist directory)
5.	Data Transactions	File I/O performance and relationship mapping without heavy RDBMS	db.json Relational Entities